Disentangling Density from Cycles in Multi-Agent Communication Topologies

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Abstract. At least seven recent multi-agent systems enforce DAG communication topologies, claiming that acyclic structure improves performance. We prove that these claims rest on a methodological confound: for connected graphs with n vertices and m edges, the cycle rank $\beta_1 = m - n + 1$, so every comparison that varies topology while varying edge count conflates cycle structure with density. To escape this identity, we introduce directed simple cycle count $\kappa(D)$ as an experimental variable that decorrelates from density—eight directed graph families at constant $(n, m) = (8, 16)$ span κ from 0 to 47. Across 240 controlled island model GA runs on OneMax, we find a significant main effect of topology on diversity (ANOVA $p < 10^{-7}$, $\eta^2 = 0.17$) with Pearson $r = -0.68$ between κ and diversity at generation 30, with the effect proving transient on OneMax. However, a supplementary NK landscape pilot shows $14\times$ amplification at $K = 6$ ($\eta^2 = 0.69$ for diversity), confirming that topology effects scale with landscape ruggedness. The existence of these effects at constant density demonstrates that the DAG hegemony rests on confounded evidence, not controlled experiment.

1 Introduction

In island model genetic algorithms, the *migration topology*—a graph whose vertices are subpopulations and whose edges determine migration routes—shapes diversity maintenance and convergence speed [19, 2]. Ring topologies maintain diversity longer than complete graphs; sparse topologies slow premature convergence [1, 18]. The open question is not *whether* topology matters but *which topological property* is operative.

Recent LLM-based multi-agent systems (MAS) have intensified this question. Seven major systems—GoAgent [5], AgentConductor [20], Graph-GRPO [4], OFA-MAS [13], G-Designer [21], ARG-Designer [12], and Dochkina et al. [7]—enforce or recommend DAG communication topologies. Against this consensus, BIGMAS [8] permits cycles and Puppeteer [6] discovers cyclic structures under optimization pressure. The field is split, with no shared framework to adjudicate.

The reason is algebraic. For any connected graph G with n vertices and m edges, the cycle rank $\beta_1(G) = m - n + 1$ [9]: cycle count and edge count differ by a constant. Since Pearson correlation is affine-invariant, $\rho(\beta_1, Y) = \rho(m, Y)$ for any outcome Y. Every comparison of "cyclic" versus "acyclic" topologies therefore simultaneously varies density. The DAG hegemony and pro-cycle findings are both potentially density effects in disguise.

We resolve this via directed graph theory. The simple directed cycle count $\kappa(D)$ is not determined by (n, m): two digraphs with $n = 8$, $m = 16$ can have 0 or 47 simple directed cycles. Across 240 runs (8 topology families, 30 seeds) at constant density, we find a significant main effect on diversity ($\eta^2 = 0.17$, $p < 10^{-7}$) with $r = -0.68$ between κ and diversity at generation 30.

The contributions are: **(1)** a theorem proving the density–cycle confound for connected undirected graphs; **(2)** directed simple cycle count $\kappa(D)$ as a density-independent experimental variable; **(3)** the first controlled experiment showing that directed cycle structure (κ) has an independent effect on island model GA dynamics; and **(4)** NK landscape results showing 14× amplification on rugged landscapes.

2 Background

2.1 Island Model Evolutionary Algorithms

In island model EAs, a population is partitioned into semi-isolated subpopulations that evolve independently and exchange individuals via migration [19]. The *migration topology* $G = (V, E)$ determines which islands may exchange migrants. The *density* of G is $|E|/|V|$, the average number of connections per island.[1] Sparse topologies slow the spread of fit individuals, maintaining diversity; dense topologies accelerate convergence but risk premature diversity loss [1, 18].

2.2 Cycle Rank and the First Betti Number

For a graph G with n vertices, m edges, and c connected components, the first Betti number is $\beta_1(G) = m - n + c$. For connected graphs ($c = 1$), $\beta_1 = m - n + 1$. Geometrically, β_1 counts independent cycles: a spanning tree uses $n - 1$ edges, and each additional edge closes exactly one independent cycle [9]. Prior work reports correlations between β_1 and fitness convergence, but as Section 3 shows, these are algebraically indistinguishable from density effects.

2.3 Directed Graphs and Simple Cycle Count

A digraph $D = (V, A)$ has arcs (ordered pairs) rather than edges. The *simple directed cycle count* $\kappa(D)$ is the number of elementary directed cycles in D. Unlike β_1, which for connected graphs is determined by n and m, $\kappa(D)$ depends on the specific arc arrangement. Enumeration uses Johnson's algorithm [10].

[1] Some authors use $|E|/\binom{|V|}{2}$; the choice does not affect the confound since both are affine in $|E|$.

Table 1. Systems enforcing the DAG constraint. All vary β_1 and density simultaneously.

System	Claim	Confound
GoAgent [5]	DAG enables tractable CIB	DAG vs. dense baseline varies β_1 and m
Graph-GRPO [4]	Learned DAGs beat complete graphs	Complete = max density and max β_1
OFA-MAS [13]	Sparse communication improves efficiency	Pruning reduces m and β_1 in lockstep
G-Designer [21]	Moderate connectivity best	No density-controlled baseline
AgentConductor [20]	DAG topology evolution for code generation	Evolves DAGs only; cyclic alternatives never tested
ARG-Designer [12]	Autoregressive DAG generation	DAG constraint baked into generative model
Dochkina et al. [7]	Sparse beats dense; capability moderates	Sparsity and acyclicity confounded in all 25K runs

3 The Density–Cycle Confound

Theorem 1 (Cycle rank–density identity). *Let $G = (V, E)$ be a connected graph with $|V| = n$ and $|E| = m$. Then $\beta_1(G) = m - n + 1$.*

Proof. Any spanning tree of G uses $n - 1$ edges; each remaining edge closes exactly one independent cycle. The result follows from the Euler characteristic $\chi = n - m = 1 - \beta_1$ for connected 1-dimensional simplicial complexes.

Corollary 1. *For connected graphs with fixed n, β_1 and m differ by the constant $n-1$. Pearson correlation is invariant under affine transformation, so $\rho(\beta_1, Y) = \rho(m, Y)$ for any outcome variable Y.*

The methodological consequence: *any experiment comparing connected topologies on a fixed node count cannot distinguish "cycles affect performance" from "edge count affects performance."* Table 1 surveys seven systems exhibiting this confound.

3.1 The Directed Escape

Theorem 1 applies to homological cycle rank. For directed graphs, the simple cycle *count* $\kappa(D)$ is combinatorial, not homological: it depends on arc arrangement, not cycle space dimension. We constructed eight families at constant $(n, m) = (8, 16)$ with κ ranging from 0 to 47. Among strongly connected members alone, κ spans 10 to 47—a 4.7× variation at strictly constant density and identical directed rank 9.

4 Experimental Design

All experiments below use the directed simple cycle count $\kappa(D)$, not the homological β_1, as the independent variable.

4.1 Graph Families

Table 2 describes the eight directed graph families, all with $n = 8$ vertices and $m = 16$ arcs (density $m/n = 2.0$ held constant). Cycle counts were verified by exhaustive enumeration via Johnson's algorithm [10].

Table 2. Eight directed graph families at constant density ($n = 8$, $m = 16$). κ = directed simple cycle count. SC = strongly connected.

Family	Description	κ	SC
`dag-layer`	3-layer forward hierarchy	0	No
`dag-wide`	Fan-out; single coordinator broadcasts	0	No
`lowcyc-1`	`dag-layer` + one feedback arc	3	No
`bidir-ring`	Forward and backward directed 8-ring	10	Yes
`two-cliques`	Two directed 4-cycles with cross-chords	14	No
`mesh-cyclic`	2×4 grid with bidirectional verticals	20	Yes
`dense-triangles`	Overlapping directed triangles + skip-2	29	Yes
`ring-skip2`	Ring $i \to i+1$ plus $i \to i+2$ (mod 8)	47	Yes

4.2 Island Model GA

Each vertex hosts one subpopulation of 50 individuals (400 total). The fitness function is OneMax on $L = 100$-bit strings: $f(\mathbf{x}) = L^{-1} \sum_{k=1}^{L} x_k \in [0, 1]$. OneMax is unimodal, chosen so that any topology effect is structural, not landscape-specific.

Selection is tournament ($k = 3$); crossover is uniform ($p = 0.5$/locus); mutation is bit-flip at rate $p_m = 1/L = 0.01$ per bit. Every 10 generations, the 5 fittest individuals on each island migrate along outgoing arcs, replacing the 5 weakest (synchronous migration). All families share density $16/8 = 2.0$. Runs last 500 generations.

4.3 Metrics and Statistical Analysis

Diversity. Mean pairwise normalised Hamming distance across all 400 individuals:

$$D(t) = \frac{1}{\binom{N}{2}} \sum_{i<j} \frac{d_H(\mathbf{x}_i, \mathbf{x}_j)}{L},$$

estimated from 20 randomly sampled pairs when $N > 20$.

Analysis. Pearson r between κ and each metric at generations 30, 60, and 100. One-way ANOVA ($\alpha = 0.05$) with topology as factor; effect size η^2; post-hoc Mann–Whitney U with Bonferroni correction. Because all families share $m = 16$, $r(\text{density}, Y) = 0$ identically. Each topology runs with 30 seeds (240 total). Implementation: Haskell; cycle counts pre-computed via Johnson's algorithm in Python.

4.4 NK Landscape Pilot

To test scaling with landscape difficulty, we ran a pilot on NK landscapes [11] ($N = 100$, adjacent neighbours, $K \in \{0, 4, 6\}$, fixed seed). Three topologies spanning κ: dag-layer (0), bidir-ring (10), ring-skip2 (47); 10 seeds per condition (90 runs).

5 Results

5.1 Topology Significantly Affects Diversity and Fitness

One-way ANOVA yields $F(7, 232) = 6.90$, $p = 1.85 \times 10^{-7}$ at generation 30; Kruskal–Wallis corroborates ($H = 38.53$, $p = 2.40 \times 10^{-6}$). Effect sizes: $\eta^2 = 0.17$ (diversity), $\eta^2 = 0.24$ (fitness)—medium-to-large. Density is constant by construction, so $r(\text{density}, Y) = 0$ identically.

5.2 Cycle Count Negatively Predicts Diversity

Figure 3 shows $\kappa(D)$ against mean diversity at generation 30. Pearson $r = -0.681$ ($r^2 = 0.46$, $p = 0.063$ on 6 df); the marginal p reflects only 8 topology-level points, while the 240-run ANOVA confirms the effect. Post-hoc Bonferroni contrasts: dag-wide ($\kappa = 0$) differs from all $\kappa \geq 10$ topologies ($p < 0.001$).

5.3 The Effect Is Transient

The temporal profile (Figures 1–4): at generation 30, $r = -0.681$, $\eta^2 = 0.17$; at generation 50, $r = 0.838$ for fitness ($p = 0.009$; higher κ slows convergence, yielding lower fitness at generation 50); by generation 100, $|r| < 0.10$, $\eta^2 = 0.002$. OneMax is unimodal, so all topologies eventually converge; κ affects the *rate* of diversity loss, not the outcome. DAG families converge fastest; ring-skip2 ($\kappa = 47$) sustains diversity longest. The bidirectional ring does not fall monotonically, suggesting cycle *length distribution* may contribute independently [17].

5.4 Topology Effect Scales with Landscape Ruggedness

Figure 5 shows η^2 for diversity increasing monotonically with K: from 0.049 ($K = 0$) to 0.445 ($K = 4$) and 0.686 ($K = 6$, both $p < 10^{-3}$) at generation 200. Fitness follows: $\eta^2 = 0.001, 0.337, 0.349$. On the rugged $K = 6$ landscape, nearly 70% of diversity variance is explained by topology—a 14× amplification over $K = 0$. The $K = 0$ control confirms these are genuine topology–landscape interactions.

High-κ topologies slow allele propagation, extending exploration. On OneMax this merely delays convergence; on rugged landscapes, sustained diversity prevents premature trapping. Ring-skip2 ($\kappa = 47$) maintains highest diversity but not highest fitness at $K = 6$—bidir-ring ($\kappa = 10$) achieves marginally better fitness, suggesting a Goldilocks effect. HERA [14] independently converges to "compact chains with strategically preserved cycles"—intermediate κ under optimization pressure.

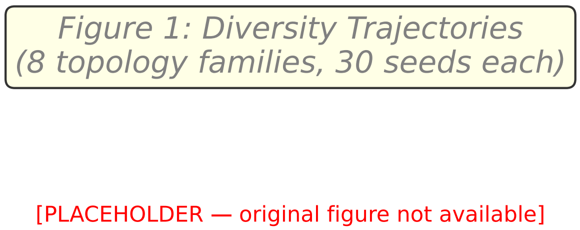

Fig. 1. Mean diversity trajectories ($\pm$ 1 SE) for 8 topology families (30 seeds each). DAG families converge fastest; ring-skip2 ($\kappa = 47$) sustains diversity longest.

6 Discussion

At constant density, $\kappa(D)$ explains 17% of diversity variance and 24% of fitness variance at generation 30—medium-to-large effects on a landscape deliberately chosen to be easy. We interpret the temporal profile as *cycle-mediated exploration*: feedback loops slow the propagation of dominant alleles, extending the period during which subpopulations maintain independent search. On OneMax, selection pressure eventually overwhelms this phase. The NK pilot confirms: η^2 rises from 0.049 ($K = 0$) to 0.686 ($K = 6$), sustained above 0.50 at generation 500 while collapsing to near zero for $K = 0$.

Diversity as the primary channel. Across both experiments, diversity is most sensitive to topology. On NK landscapes, diversity $\eta^2 = 0.686$ while best-fitness $\eta^2 < 0.10$. Cycles operate through population-level diversity maintenance rather than direct fitness improvement. Practitioners should monitor diversity, not just fitness, when evaluating communication topologies.

The implications for MAS design are immediate. The systems in Table 1 concluded DAGs outperform, but by Theorem 1 their experiments compared sparse to dense. The correct conclusion is that *sparser communication is cheaper*, not that cycles are harmful.

Figure 6 illustrates the confound concretely. We ran the same island model GA on 13 cubic symmetric graphs from the Foster census [3] ($n = 4$ to 30, 30 seeds each). Because every 3-regular graph on n vertices has $m = 3n/2$ edges, the cycle rank $\beta_1 = n/2 + 1$, and the density $m/n = 3/2$—all three quantities are affine in n, hence perfectly collinear. The left panel plots diversity against n

Figure 2: Pearson r(kappa, diversity) over generations

[PLACEHOLDER — original figure not available]

Fig. 2. Pearson $r(\kappa, \text{diversity})$ over generations. Peak at gen. 20–50; decay by gen. 100.

(increasing trend); the right plots the *same data* against density $m/\binom{n}{2}$ (decreasing trend). Both narratives are the same regression—the trap Theorem 1 predicts. Only constant-density experiments break the degeneracy. As a positive control, the Dodecahedron and Desargues graphs share $n = 20$, $\beta_1 = 11$, density $3/19$; Mann–Whitney finds no significant difference ($p = 0.615$), confirming that graphs matched on all confounded variables produce indistinguishable dynamics.

Limitations. First, $n = 8$ islands is small; scaling experiments on larger graph families are needed (the Foster sweep uses up to $n = 30$ but cannot escape the density confound within the 3-regular family). Second, the Pearson correlation is marginal ($p = 0.063$) on 8 topology-level points, though significant at generation 50 ($r = 0.838$, $p = 0.009$) and corroborated by the 240-run ANOVA ($p < 10^{-7}$). Third, the NK pilot uses only 3 of 8 topologies and 10 seeds; a full-factorial experiment is needed for generality.

7 Related Work

Learned and adaptive topologies. Puppeteer [6] trains an RL orchestrator that discovers "compact, cyclic reasoning structures" on hard tasks, outperforming DAG-constrained MacNet [16]. Cycles *emerge* under optimization pressure, but no topological metric (β_1, κ) is computed and no ablation separates cycles from density. By Theorem 1, their evidence is exactly the confound our experiment resolves. HERA [14] is the first system to explicitly track cycle count during topology evolution, converging to "compact chains with strategically preserved cycles"—intermediate κ under optimization pressure. Our theorem explains why this works.

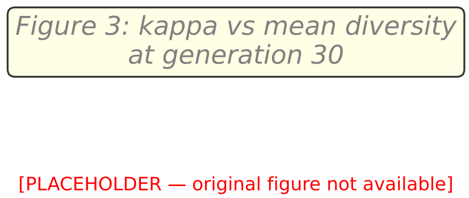

Fig. 3. κ vs. mean diversity at gen. 30. $r = -0.681$ ($p = 0.063$, 6 df).

Large-scale studies. Lynch et al. [15] show that graph-parameterised clock systems constrain representable behaviours to paths in G: DAGs permit only acyclic execution. Dochkina et al. [7] (25K+ runs) find sparse/acyclic outperforms dense/cyclic with capability-moderated crossover. By Theorem 1, this is a density comparison; our experiment provides the controlled condition.

Cycle structure. CycRak [17] shows short-cycle centrality outperforms eigenvector centrality for diffusion, suggesting cycle *length distribution* governs propagation—consistent with our non-monotonic bidir-ring result.

8 Conclusion

For connected graphs with fixed vertex count, cycle rank β_1 and edge count m differ by a constant (Theorem 1). Every prior comparison of cyclic versus acyclic topologies is therefore confounded. We escape this by moving to directed graphs, where $\kappa(D)$ is not determined by (n, m). Across 240 runs at constant density, κ predicts diversity ($\eta^2 = 0.17$, $p < 10^{-7}$) and fitness ($\eta^2 = 0.24$). The effect is transient on OneMax but a 14× amplification at NK $K = 6$ confirms scaling with landscape difficulty. Future work should pursue full-factorial NK experiments, κ–capability interaction, and κ as a practical topology design heuristic.

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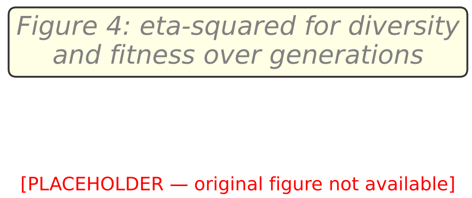

Fig. 4. η^2 for diversity and fitness over generations. Peak near gen. 30; decay by gen. 100.

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Figure 6: eta-squared by metric across NK ruggedness (K)

[PLACEHOLDER — original figure not available]

Fig. 5. η^2 by metric across NK ruggedness (K) at gen. 200. Topology explains $< 5\%$ at $K = 0$ but $\sim$70% of diversity at $K = 6$. Dashed: Cohen's thresholds. $^{**}p < 0.01$; $^{***}p < 0.001$.

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Figure 5: Same data, opposite narrative
(Foster census cubic symmetric graphs)

[PLACEHOLDER — original figure not available]

Fig. 6. Same data, opposite narrative. Left: diversity vs. n. Right: diversity vs. density $m/\binom{n}{2}$. 13 Foster census cubic symmetric graphs ($n = 4$–30, 30 seeds). For 3-regular graphs, β_1, m, n are collinear—causal attribution is underdetermined.